SAI LIKHITHA PULAGAM

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SUMMARY

- Data Engineer with nearly **4** years of experience designing, building, and optimizing scalable ETL/ELT pipelines, relational and NoSQL data models, and enterprise data governance frameworks across **consulting**, **healthcare**, and **telecom industries**.
- Proficient in **Python**, **SQL**, and **MongoDB** — with hands-on expertise in document schema design, indexing, partitioning, and data sharding for high-performance operational and analytical workloads across relational and NoSQL database environments.
- Experienced in end-to-end data engineering across **AWS** and **GCP** — from multi-source ingestion and real-time replication (HVR, CDC) to warehouse performance optimization, data quality validation, and compliance-ready pipeline delivery.
- Adept at working independently on complex data engineering problems and collaborating cross-functionally in **Agile** Scrum environments — translating business requirements into secure, reliable, and scalable data solutions that drive operational decision-making.

TECHNICAL SKILLS

- **Modern Data Stack & Orchestration:** dbt, HVR, Apache Airflow, Azure Data Factory (ADF), Informatica, SSIS, CI/CD, Systems Thinking, Analytics Engineering
- **Programming Languages & Scripting:** Python, SQL, Scala, Bash
- **Big Data & Data Processing:** Apache Spark, Hadoop, Delta Lake, Kafka
- **Cloud Platforms & Services: Azure (**Data Factory, Synapse Analytics, Databricks, Blob Storage, SQL Database, DevOps), AWS **(**S3, EC2, Lambda, Glue, Redshift)
- **Databases/Warehousing:** Snowflake, Redshift, Azure Synapse, MySQL, PostgreSQL, SQL Server, MongoDB
- **AI & Machine Learning Libraries:** Scikit-learn, Pandas, NumPy, GenAI (basic), NLP (basic)
- **Data Modeling:** Dimensional Modeling, Star/Snowflake Schemas, ELT/ETL Development, Data Pipeline Design, Data Quality Checks
- **Data Visualization & Monitoring:** Power BI, Tableau, Excel, Custom Dashboards
- **Tools & DevOps:** Git, Azure DevOps, Docker, Kubernetes, Jenkins, Great Expectations

PROFESSIONAL EXPERIENCE

Data Engineer | Verizon — **Mar 2025 – Present**

- Built and optimized scalable **ELT** data models and pipeline schemas on **Google BigQuery** — supporting end-to-end migration strategies for **FPAS** and financial systems, processing **100M+** daily records with data lineage traceability, source-to-target mappings, and full compliance with enterprise governance standards across multiple business domains.
- Refactored and optimized complex **Teradata TPT** scripts and **SQL stored procedures** into BigQuery-compatible queries — improving query performance by **35%** and reducing pipeline runtime by **65%,** while preserving schema consistency, data accuracy, and business logic integrity across financial datasets.
- Developed and optimized **Apache Airflow DAGs** using Python to implement fault-tolerant data transformation and processing pipelines across development, staging, and production environments — integrating 20+ upstream data sources and ensuring reliable delivery of financial reporting data for downstream stakeholders.
- Integrated **MongoDB** as an operational data store alongside BigQuery — designing document schemas and collection structures for real-time transactional data, implementing indexing, partitioning, and data sharding strategies to optimize read/write performance and horizontal scalability, and building Python-based ingestion pipelines to sync operational records with BigQuery for enterprise analytics and reporting.
- Built incremental data migration frameworks using BigQuery MERGE and SQL scripting — synchronizing 50+ datasets, reducing data latency from 24 hours to 2 hours, and performing automated Python and SQL reconciliation to resolve 100% of record-count discrepancies before production promotion.
- Executed **Historical Data Load (HDL) processes** to migrate 5+ years of legacy financial data across **200+ datasets** totaling **40+ TB** — conducting data analysis and performance tuning throughout to validate accuracy, track replication priorities, and eliminate analyst dependency on legacy Teradata systems.
- Implemented BigQuery data governance using policy tags, column-level security, and **PII masking** to protect 100M+ sensitive financial records — enforcing data security and compliance best practices, applying HIPAA-aligned access controls, and delivering audit-ready analytics environments for enterprise reporting.
- Optimized BigQuery warehouse performance through partitioning, clustering, and query tuning — applying indexing and data sharding strategies to reduce compute costs by ~30% while improving throughput for 300+ downstream analytics and reporting users across financial business domains.
- Applied **CI/CD** best practices within **Agile** Scrum delivery cycles — participating in sprint planning, daily stand-ups, and retrospectives to drive incremental feature delivery, managing pull request workflows, conducting peer code reviews, and implementing automated data quality gates that reduced production defects by ~40%, while maintaining documentation for transformations, business rules, and compliance audits.

Environment: Google Cloud Platform (BigQuery, Dataflow, Cloud Composer, Pub/Sub, Dataproc, Cloud Storage), Apache Airflow, PySpark, Python, SQL, Teradata, MongoDB, Oracle, MySQL, Looker, Great Expectations, Jenkins, Git, Avro, Parquet, JSON, Agile

Data Analyst | Kent State University | OH — **Aug 2023 – Dec 2024**

- Developed and executed SQL queries — including **joins**, **aggregations**, and **window functions** — and **Python (Pandas, NumPy)** scripts to perform exploratory data analysis on university datasets, reducing report generation time by **~35%** and identifying trends in student performance, enrollment, and retention to support data-driven institutional decision-making.
- Developed and delivered **10+** interactive dashboards using **Power BI** and **Tableau** — analyzing enrollment and academic performance data to transform raw institutional records into actionable reporting for administrative and academic leadership decision-making.
- Collaborated with institutional research and academic departments to streamline data collection workflows — ensuring data consistency, integrity, and compliance across multiple sources, applying data security best practices to protect sensitive student records.
- Built automated **ETL pipelines** and data validation scripts in Python (Pandas, NumPy) processing **500K+ academic records**, implementing data quality checks across university datasets, and reducing manual data preparation effort by 40%, improving reliability for downstream reporting and analysis

Environment: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Power BI, Tableau, Jupyter Notebook, Excel

Data Engineer | LTI Mind Tree | IN **July 2021 – Dec 2022**

- Built and maintained scalable **ETL/ELT** pipelines and dimensional data warehouse **schemas (Star and Snowflake)** within Agile Scrum delivery cycles — migrating 500M+ financial records from **legacy systems**, **Teradata**, and **Informatica** workflows using **Python**, **SQL**, **AWS Glue**, **PySpark,** and Snowflake, ensuring secure, fault-tolerant execution across batch and real-time enterprise analytics environments.
- Engineered **multi-source data ingestion** and **real-time replication** pipelines using **HVR** from **MongoDB**, **MySQL**, and **NoSQL** systems as the source layer feeding enterprise migration workflows — processing 2.3M+ records daily using **PySpark** and **AWS EMR**, enabling change data capture (CDC), real-time dashboards, and BI reporting for downstream business stakeholders.
- Developed automated data validation frameworks **using Python (Pandas, NumPy)** with unit tests and quality rules to verify accuracy and consistency of ingested records across pipelines — ensuring reliability of millions of financial records supporting fraud detection and regulatory reporting, while reducing manual data preparation effort by **30+ hours weekly**.
- Developed and optimized complex **SQL stored procedures, T-SQL functions**, **views**, and **window functions** to transform validated datasets into reporting-ready structures — reducing reporting latency from hours to minutes and improving query performance by **40%+** across enterprise financial and operational datasets.
- Designed and deployed -dimensional data warehouse schemas in Snowflake — applying **indexing**, **partitioning**, and **data sharding** strategies to reduce BI query times by 40% and support high-performance analytics access for 100+ business users.
- Enforced enterprise data security and compliance standards across the full data pipeline lifecycle — implementing audit controls, column-level access management, and pipeline validation to protect sensitive operational datasets, delivering secure, governed, and audit-ready data environments across regulated infrastructure.

Environment: AWS Glue, S3, Redshift, Snowflake, dbt, HVR, Spark, Python, SQL, Tableau, Power BI, Git

PROJECTS

YT Sentinel — YouTube Analytics Pipeline | Mango Mass Media
Professional Project | GCP Cloud-Native Data Engineering

- Built an end-to-end real-time data ingestion and processing pipeline using **Python and Google Cloud Run** to collect and classify YouTube comments via the YouTube Data API — implementing a sharding strategy across **30 shards via Rendezvous Hashing** and a **Pub/Sub messaging** layer with three topic streams to decouple microservices, processing **300K+ daily quota units** with fault-tolerant, event-driven data flow across hot, active, cold, and backfill tiers.
- Designed **NoSQL document schemas in Firestore** for real-time operational storage of video metadata and comments — integrating **Vertex AI** for sentiment classification before warehousing to Big Query and developing SQL-based warehouse schemas with watermark deduplication and SLA enforcement (delete ≤30min, reply ≤90min) to ensure data accuracy and timeliness for downstream analytics.
- Enforced enterprise data security using **GCP Secret Manager** for API token and credential management — implementing pipeline validation, health monitoring via **Cloud Monitoring dashboards**, and access controls across all Cloud Run microservices to deliver a secure, governed, and audit-ready analytics platform.

Environment: Python, SQL, Google Cloud Run, Pub/Sub, BigQuery, Firestore, Vertex AI, Cloud Scheduler, Cloud Storage, Memory store, Secret Manager, Cloud Monitoring, YouTube Data API, Git

Customer Churn Prediction & Segmentation
AI/ML Data Engineering Project | 2024 · Python, Pandas, scikit-learn, XGBoost, Snowflake, Power BI

- Built automated data preparation and ETL pipelines using **Python and Pandas** to extract, transform, and unify customer datasets from **Snowflake** — creating a clean, analytics-ready data model for churn scoring and behavioral segmentation across multiple customer cohorts.
- Trained and validated **XGBoost and logistic regression models** to estimate customer churn risk — applying clustering techniques to segment customers by behavior and usage patterns, supporting targeted retention and marketing engagement strategies.
- Automated model scoring and reporting pipelines **using Python and Snowflake** — enabling repeatable, scheduled churn analysis and delivering Power BI dashboards combining churn scores and customer segments to visualize risk trends and distribution for business stakeholders.

Environment: Python, Pandas, scikit-learn, XGBoost, Snowflake, SQL, Power BI, Git

CERTIFICATIONS

- MongoDB Associate Developer – **MongoBD University**
- AWS Certified Data Engineer **— Associate | AWS**
- SnowPro Core Certification **— Snowflake**

EDUCATION

Master of Science in Computer Science
Kent State University, OH

Bachelor of Technology in Computer Science
JNTU Kakinada, India